

6. Saturated and Unsaturated Fats

Saturated and unsaturated fats are found in different amounts in different foods.

Unsaturated fats are often described as ‘good fats’. When used to replace saturated fats in the diet they help lower cholesterol, one of the risk factors linked to heart disease.

While full-fat dairy products and meat contain large amounts of saturated fats, sources of unsaturated fats include nuts, seeds and vegetable oils.

Saturated and unsaturated fats differ in their chemical structures. Unsaturated fats contain double bonds between the carbon atoms within their molecular structure, whereas saturated fats have no double bonds.



The label on the bottle of a new brand of cooking oil claims that it is better than other cooking oils available on the market because it contains only **15% saturated fat**.

Scientists decided to investigate this claim.

They used bromine water to compare the percentage unsaturation of this new brand of cooking oil with four other vegetable oils, **W**, **X**, **Y** and **Z**. They also tested a known alkane and alkene.

They measured the volume of bromine water that could be added to each oil sample before the bromine water colour remained. Their results are shown below.

| Sample        | Mean volume of bromine water added (cm <sup>3</sup> ) | Unsaturation (%) | Saturation (%) |
|---------------|-------------------------------------------------------|------------------|----------------|
| oil <b>W</b>  | 20.4                                                  | 50               | 50             |
| oil <b>X</b>  | 12.0                                                  | 30               | 70             |
| oil <b>Y</b>  | 24.1                                                  | 60               | 40             |
| oil <b>Z</b>  | 16.3                                                  | 40               | 60             |
| new brand oil | 30.1                                                  | ?                | ?              |
| alkane        | 0.1                                                   | -                | -              |
| alkene        | 40.0                                                  | -                | -              |



Examiner  
only

(a) Which **one** of these statements best describes the oils tested? Tick (✓) the correct answer. [1]

the oils contain saturated fats only

☐

the oils contain unsaturated fats only

☐

the oils contain both saturated and unsaturated fats

☐

it is not possible to tell whether the oils contain saturated or unsaturated fats

☐

(b) Explain why the first drops of bromine water that are added to each of the oil samples are decolourised. [2]

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(c) The label states that the new oil contains only 15 % saturated fat. Use the results to show whether this statement is correct. [3]

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(d) Explain why these results might persuade many consumers to use this new brand of oil in preference to the other available brands. [2]

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Examiner  
only

6. Discuss what is meant by isomerism in alkanes and alkenes using compounds with the molecular formulae  $C_4H_{10}$  and  $C_4H_8$  to illustrate your answer. [6 QER]

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END OF PAPER

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### FORMULAE FOR SOME COMMON IONS

| POSITIVE IONS |                  | NEGATIVE IONS |                    |
|---------------|------------------|---------------|--------------------|
| Name          | Formula          | Name          | Formula            |
| aluminium     | $\text{Al}^{3+}$ | bromide       | $\text{Br}^-$      |
| ammonium      | $\text{NH}_4^+$  | carbonate     | $\text{CO}_3^{2-}$ |
| barium        | $\text{Ba}^{2+}$ | chloride      | $\text{Cl}^-$      |
| calcium       | $\text{Ca}^{2+}$ | fluoride      | $\text{F}^-$       |
| copper(II)    | $\text{Cu}^{2+}$ | hydroxide     | $\text{OH}^-$      |
| hydrogen      | $\text{H}^+$     | iodide        | $\text{I}^-$       |
| iron(II)      | $\text{Fe}^{2+}$ | nitrate       | $\text{NO}_3^-$    |
| iron(III)     | $\text{Fe}^{3+}$ | oxide         | $\text{O}^{2-}$    |
| lithium       | $\text{Li}^+$    | sulfate       | $\text{SO}_4^{2-}$ |
| magnesium     | $\text{Mg}^{2+}$ |               |                    |
| nickel        | $\text{Ni}^{2+}$ |               |                    |
| potassium     | $\text{K}^+$     |               |                    |
| silver        | $\text{Ag}^+$    |               |                    |
| sodium        | $\text{Na}^+$    |               |                    |
| zinc          | $\text{Zn}^{2+}$ |               |                    |



2  
1

## Group

3

4

5

9

2

0

|                            |
|----------------------------|
| $^1_1\text{H}$<br>Hydrogen |
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|                                    |                                    |
|------------------------------------|------------------------------------|
| 7<br><b>Li</b><br>Lithium<br>3     | 9<br><b>Be</b><br>Beryllium<br>4   |
| 23<br><b>Na</b><br>Sodium<br>11    | 24<br><b>Mg</b><br>Magnesium<br>12 |
| 39<br><b>K</b><br>Potassium<br>19  | 40<br><b>Ca</b><br>Calcium<br>20   |
| 86<br><b>Rb</b><br>Rubidium<br>37  | 88<br><b>Sr</b><br>Strontium<br>38 |
| 133<br><b>Cs</b><br>Caesium<br>55  | 137<br><b>Ba</b><br>Barium<br>56   |
| 223<br><b>Fr</b><br>Francium<br>87 | 226<br><b>Ra</b><br>Radium<br>88   |

|           |           |          |            |            |           |         |           |        |         |     |
|-----------|-----------|----------|------------|------------|-----------|---------|-----------|--------|---------|-----|
| 45        | Sc        | 48       | 51         | 52         | 55        | 56      | 59        | 59     | 63.5    | 65  |
| Scandium  | Titanium  | Vanadium | Chromium   | Manganese  | Iron      | Cobalt  | Nickel    | Copper | Zinc    |     |
| 21        | 22        | 23       | 24         | 25         | 26        | 27      | 28        | 29     | 30      |     |
|           |           |          |            |            |           |         |           |        |         |     |
| 89        | Y         | 91       | 93         | 96         | 99        | 101     | 103       | 106    | 108     | 112 |
| Yttrium   | Zirconium | Niobium  | Molybdenum | Technetium | Ruthenium | Rhodium | Palladium | Silver | Cadmium |     |
| 39        | 40        | 41       | 42         | 43         | 44        | 45      | 46        | 47     | 48      |     |
|           |           |          |            |            |           |         |           |        |         |     |
| 139       | La        | 179      | 181        | 184        | 186       | 190     | 192       | 195    | 197     | 201 |
| Lanthanum | Hafnium   | Tantalum | Tungsten   | Rhenium    | Osmium    | Iridium | Platinum  | Gold   | Mercury |     |
| 57        | 72        | 73       | 74         | 75         | 76        | 77      | 78        | 79     | 80      |     |

|                             |                             |                             |                              |                              |                           |
|-----------------------------|-----------------------------|-----------------------------|------------------------------|------------------------------|---------------------------|
| 11<br>B<br>Boron<br>5       | 12<br>C<br>Carbon<br>6      | 14<br>N<br>Nitrogen<br>7    | 16<br>O<br>Oxygen<br>8       | 19<br>F<br>Fluorine<br>9     | 20<br>Ne<br>Neon<br>10    |
| 27<br>Al<br>Aluminium<br>13 | 28<br>Si<br>Silicon<br>14   | 31<br>P<br>Phosphorus<br>15 | 32<br>S<br>Sulfur<br>16      | 35.5<br>Cl<br>Chlorine<br>17 | 40<br>Ar<br>Argon<br>18   |
| 70<br>Ga<br>Gallium<br>31   | 73<br>Ge<br>Germanium<br>32 | 75<br>As<br>Arsenic<br>33   | 79<br>Se<br>Selenium<br>34   | 80<br>Br<br>Bromine<br>35    | 84<br>Kr<br>Krypton<br>36 |
| 115<br>In<br>Indium<br>49   | 119<br>Sn<br>Tin<br>50      | 122<br>Sb<br>Antimony<br>51 | 128<br>Te<br>Tellurium<br>52 | 127<br>I<br>Iodine<br>53     | 131<br>Xe<br>Xenon<br>54  |
| 204<br>Tl<br>Thallium<br>81 | 207<br>Pb<br>Lead<br>82     | 209<br>Bi<br>Bismuth<br>83  | 210<br>Po<br>Polonium<br>84  | 210<br>At<br>Astatine<br>85  | 222<br>Rn<br>Radon<br>86  |

## Key

| $A_r$ | Symbol | Name | $Z$ |
|-------|--------|------|-----|
|-------|--------|------|-----|

relative atomic mass

atomic number

|               |               |                  |
|---------------|---------------|------------------|
| Surname       | Centre Number | Candidate Number |
| First name(s) |               | 0                |



GCSE

3430UE0-1



FRIDAY, 27 MAY 2022 – MORNING

SCIENCE (Double Award)

Unit 5 – CHEMISTRY 2

HIGHER TIER

1 hour 15 minutes

| For Examiner's use only |              |              |
|-------------------------|--------------|--------------|
| Question                | Maximum Mark | Mark Awarded |
| 1.                      | 7            |              |
| 2.                      | 8            |              |
| 3.                      | 9            |              |
| 4.                      | 15           |              |
| 5.                      | 8            |              |
| 6.                      | 6            |              |
| 7.                      | 7            |              |
| Total                   | 60           |              |

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 7(a) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



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6. It is well known that plastic bags are a major source of litter. These bags can exist for a very long time, ranging from tens of years in the natural environment up to a thousand years in landfill.

The use of plastic bags is now banned in some countries. These countries commonly use bags made from paper or cotton, which do not cause the same litter problem as plastic bags.

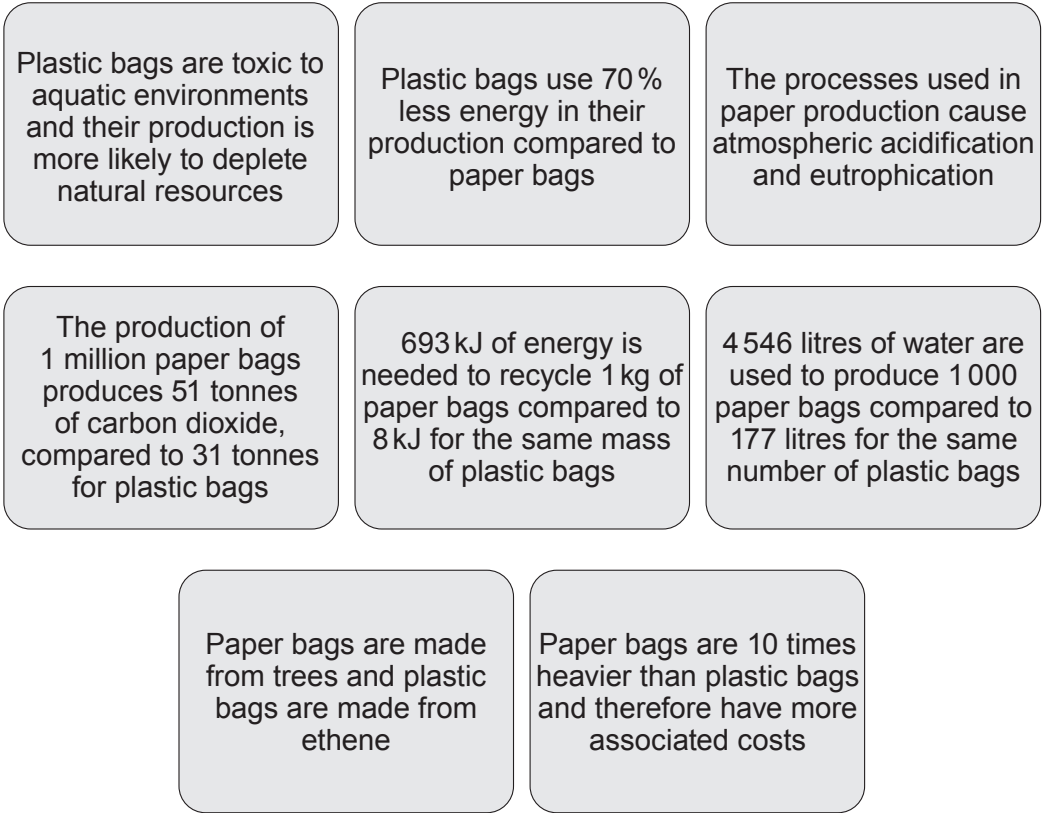
Paper or plastic?



Most people assume that a paper bag is better for the environment, however because paper bags are almost ten times heavier than plastic bags, they produce a greater mass of waste. On the other hand, because paper bags are biodegradable they do not lead to the same problem with litter as plastic bags.

When deciding whether paper is more suitable than plastic for making shopping bags, information from life cycle assessments (LCAs) of both materials must be taken into consideration.

Some of the findings from LCAs of paper and plastic bags are shown below.



On reading these findings it is important to consider all factors relating to sustainability when deciding on the most suitable material to make shopping bags. We should not just be thinking about what happens when we throw them away!



Examiner  
only

- (a) Use the information in the passage to compare the mass of waste and volume of litter generated from the use of plastic and paper bags. [2]

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- (b) Use the information from the LCAs to tick (✓) whether each of the factors supports or opposes the use of paper bags in preference to plastic bags. [3]

|                                          | Supports use of<br>paper bags | Opposes use of<br>paper bags |
|------------------------------------------|-------------------------------|------------------------------|
| Impact of waste on marine life           | <input type="checkbox"/>      | <input type="checkbox"/>     |
| Water consumption in production          | <input type="checkbox"/>      | <input type="checkbox"/>     |
| Energy used in production                | <input type="checkbox"/>      | <input type="checkbox"/>     |
| Carbon footprint generated in production | <input type="checkbox"/>      | <input type="checkbox"/>     |
| Energy used in recycling                 | <input type="checkbox"/>      | <input type="checkbox"/>     |
| Cost of transporting waste               | <input type="checkbox"/>      | <input type="checkbox"/>     |

- (c) The use of which type of bag is more likely to be linked to the erosion of limestone buildings?  
Give a reason for your answer. [1]

Type of bag .....

Reason .....

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6. Hydrogels are macromolecules, meaning they contain very large numbers of atoms.

They are polymers made from hydrophilic polymer chains. This means that they readily mix with water.

Cross-links between the polymer chains enable hydrogels to form their three-dimensional shape.

The cross-links that join the individual polymer chains together are formed by small reactive molecules that contain one or more double bonds.



The interesting feature of hydrogels is their ability to swell and shrink, depending on their surroundings. When in contact with water, hydrogels are able to absorb water and swell to many times their original size and weight, whilst maintaining the same shape.

Being able to absorb water like this has enabled hydrogels to be used in a wide range of applications. For many of these applications it is important to understand how the properties of hydrogels change with temperature.

**Table 1** and **Table 2** show how temperature affects the properties of hydrogel beads in water over time.

| Time (hours) | Mass of bead (g) |       |       |
|--------------|------------------|-------|-------|
|              | 10 °C            | 20 °C | 40 °C |
| 0            | 0.035            | 0.035 | 0.035 |
| 2            | 1.305            | 2.980 | 4.280 |
| 4            | 2.512            | 3.681 | 5.904 |
| 6            | 3.613            | 4.389 | 6.760 |
| 8            | 4.298            | 5.195 | 8.101 |
| 10           | 5.120            | 5.572 | 8.101 |
| 12           | 5.400            | 5.980 | 8.101 |

**Table 1 – the effect of temperature on the mass of hydrogel beads**

| Time (hours) | Diameter of bead (mm) |       |       |
|--------------|-----------------------|-------|-------|
|              | 10 °C                 | 20 °C | 40 °C |
| 0            | 3.56                  | 3.56  | 3.56  |
| 2            | 10.30                 | 13.00 | 18.10 |
| 4            | 13.40                 | 16.50 | 21.70 |
| 6            | 16.20                 | 18.50 | 23.60 |
| 8            | 17.80                 | 19.60 | 25.24 |
| 10           | 19.50                 | 20.80 | 25.24 |
| 12           | 20.50                 | 22.00 | 25.24 |

**Table 2 – the effect of temperature on the diameter of hydrogel beads**



Examiner  
only

(a) Tick (✓) the statement that **best** describes the molecules that form the cross-links within hydrogels. [1]

- they are macromolecules ☐
- they are unsaturated ☐
- they contain hydrophilic groups ☐
- they are polymers ☐

(b) Give the phrase in the passage that shows the change the hydrogel undergoes is a physical rather than a chemical change. [1]

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(c) Calculate the percentage increase in **mass** of a hydrogel bead when placed in water at 20 °C for 6 hours. [2]

Percentage increase = ..... %

(d) Write a conclusion for the effect of **temperature** on the diameter of hydrogel beads in water over time. [2]

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